EXACT LAPLACE REPRESENTATION FOR A SUBORDINATOR AT INVERSE-INTEGRAL TIME

BÉNÉDICTE HAAS, BASTIEN MALLEIN, AND BÉNÉDICTE HAAS, BASTIEN MALLEIN

Editorial note. Only the designated primary theorem counts. Supporting named results and other statements are uncounted. Focused reading locates and understands the published author proof; it is not a stepwise independent proof verification.

Allowed standard prerequisites (editorial).

- Levy process fluctuation and strong Markov theory. Inverse exponent is a subordinator exponent and its Levy measure is the excursion lifetime measure. The PDE and resulting inverse-integral-time Laplace formula remain local obligations.
- Fubini, dominated convergence and Laplace inversion. Basic exchange and inversion tools only; source proves relevant integrability, mixed derivatives and initial-value decomposition.

0.1. **Main results.** Motivated by the previous discussion, we consider now a generic subordinator ξ, with Laplace exponent

$$\phi(q) = -\log \mathbb{E}\left[e^{-q\xi_1}\right] = \kappa + cq + \int_0^\infty (1 - e^{-qx})\pi(\mathrm{d}x)$$

where $\kappa \geq 0$, $c \geq 0$ and π is a measure on $(0,\infty)$ such that $\int_0^\infty (1 \wedge x)\pi(\mathrm{d}x) < \infty$. We refer to π as the Lévy measure of ξ, c its drift and κ its death rate. We define then the time change ρ by

$$\rho(t) := \inf\left\{r \geq 0 : \int_0^r \xi_u \mathrm{d}u \geq t\right\}, \quad t \geq 0,$$

with the convention $\inf \emptyset = \infty$. Our first main result gives the following exact expression for the Laplace transform of $\xi_{\rho(t)}$, for any $t \geq 0$.

Theorem 0.1. *Let $\Phi(q) = \int_0^q \phi(s)\mathrm{d}s$, $q \geq 0$ and $\Phi^{-1} : [0,\infty) \mapsto [0,\infty)$ be the inverse of Φ. Then the function Φ^{-1} is the Laplace exponent of a subordinator, with a Lévy measure that we denote L, and for all $q > 0$ and $t \geq 0$, we have*

$$\mathbb{E}\left[e^{-q\xi_{\rho(t)}}\right] = \phi(q) \int_0^\infty e^{-\Phi(q)x - \frac{t}{x}} x L(\mathrm{d}x). \tag{0.1}$$

1. The time-changed subordinator ξ_ρ and its Laplace exponent

In this and the next section, we work with ξ a generic subordinator with Laplace exponent

$$\phi(q) = \kappa + cq + \int_0^\infty (1 - \exp(-qx))\,\pi(\mathrm{d}x), \quad q \geq 0$$

where $\kappa \geq 0$, $c \geq 0$ and π is a measure on $(0,\infty)$ such that $\int_0^\infty (1 \wedge x)\pi(\mathrm{d}x) < \infty$. We assume throughout that $\pi(0,\infty) > 0$ and $\xi_0 = 0$. We recall that $\rho(t)$ is the stopping time defined by

$$\rho(t) = \inf\left\{u \geq 0 : \int_0^u \xi_r \mathrm{d}r > t\right\} \text{ for all } t \geq 0. \tag{1.1}$$

Observe that $\rho(0) = \inf\{u \geq 0 : \xi_u > 0\}$ is either null almost surely, if $c > 0$ or $\pi((0,\infty)) = \infty$, or is strictly positive almost surely, when $c = 0$ and π is finite. In the latter case, it equals the first jump time of ξ, therefore $\xi_{\rho(0)}$ is equal to the first jump of ξ. Observe also that

$$\int_0^{\rho(t)} \xi_r \mathrm{d}r = t \quad \text{for all} \quad t < \int_0^{\mathbf{e}(\kappa)} \xi_r \mathrm{d}r,$$

Date: ahl:10.5802/ahl.234:published-2025; CC BY 4.0.

1.1. **Definition and first properties of the measure L.** We recall from the statement of Theorem 0.1 that Φ is the primitive of ϕ satisfying $\Phi(0) = 0$, i.e.

$$\Phi(q) = \kappa q + \frac{c}{2}q^2 + \int_{\mathbb{R}_+} \left(q - \frac{(1 - \exp(-qx))}{x} \right) \pi(\mathrm{d}x) \tag{1.2}$$
$$= \kappa q + \frac{c}{2}q^2 + q\pi([1,\infty)) + \int_0^\infty \left(qx\mathbb{1}_{\{x<1\}} - (1 - \exp(-qx)) \right) \frac{\pi(\mathrm{d}x)}{x}.$$

A key remark is that Φ is the Laplace transform of a Lévy process with diffusion coefficient c, drift $\kappa+\pi([1,\infty))$ and a Lévy measure given by the image of $\pi(\mathrm{d}x)/x$ by $x \mapsto -x$. We denote X such a process, satisfying

$$\mathbb{E}\left[e^{qX_t}\right] = e^{t\Phi(q)} \quad \text{for all } q, t \geq 0.$$

This class of processes, usually called *spectrally negative Lévy processes*, has remarkable properties and has been studied extensively. We refer to Bertoin's book [Ber98, Chapter VII] for background. Here, we note that either X oscillates ($\limsup_{t\to\infty} X_t = -\liminf_{t\to\infty} X_t = +\infty$) when $\kappa = 0$, or tends to $+\infty$ ($\lim_{t\to\infty} X_t = +\infty$) when $\kappa > 0$, using that $\Phi'(0) = \phi(0) = \kappa$. It is well known that $\Phi^{-1} : [0,\infty) \mapsto [0,\infty)$ is the Laplace exponent of a subordinator. More precisely,

Lemma 1.1 ([Ber98, Theorem 1, Chapter VII])**.** *The process σ defined for $t \geq 0$ by*

$$\sigma_t = \inf\{s > 0 : X_s > t\} = \inf\{u > 0 : S_u > t\},$$

where $S_t = \sup_{s\leq t} X_s$, is a subordinator with Laplace exponent Φ^{-1}. Moreover S is a local time at 0 of the reflected process $S - X$. Therefore, if we let L denote the Lévy measure of σ, we have

$$L(\mathrm{d}x) = n(\zeta \in \mathrm{d}x)$$

where n denotes the excursion measure away from 0 of $S - X$ and ζ the lifetime of an excursion.

Since $\Phi^{-1}(0) = 0$ and, as $q \to \infty$, $\Phi^{-1}(q)/q$ converges to $(\kappa + \pi(0,\infty))^{-1}$ when $c = 0$ and to 0 otherwise, the subordinator σ has no killing term and a drift equal to $\mathbb{1}_{\{c=0\}}(\kappa+\pi(0,\infty))^{-1}$. Consequently the function Φ^{-1} can be written for all $q \geq 0$ as

$$\Phi^{-1}(q) = \frac{q\mathbb{1}_{\{c=0\}}}{\kappa + \pi(0,\infty)} + \int_0^\infty (1 - \exp(-qx))\, L(\mathrm{d}x). \tag{1.3}$$

1.2. **Proof of Theorem 0.1.** We now prove Theorem 0.1. In that aim we first introduce, for all $t \geq 0, q > 0$,

$$f(t,q) := (\Phi^{-1})'(q)\mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right] \tag{1.4}$$

and observe that this bivariate function f is the solution to a simple PDE.

Lemma 1.2. *The function f is a strong solution to the equation*

$$\partial_q\partial_t f = \partial_t\partial_q f = f. \tag{1.5}$$

Additionally, the function $t \in [0,\infty) \mapsto f(t,q)$ is $\mathcal{C}^1$ on $(0,\infty)$ and continuous at 0, with $f(0,q) = \int_0^\infty xe^{-qx}L(\mathrm{d}x)$, for all $q > 0$.

In the proof of this lemma, we will note some regularity properties of f that are a bit more precise than those stated here. Moreover, we will see further with Lemma 1.4 that f is infinitely differentiable on $(0,\infty)^2$.

Instead of using this approach based on the infinitesimal generator, we prefer to use a more elementary method, which consists in rewriting the function f as follows.

Lemma 1.3. *For all $t \geq 0$ and $q > 0$, the function f defined in* (1.4) *satisfies*

$$f(t,q) = \int_t^\infty \mathbb{E}\left[\frac{e^{-\Phi^{-1}(q)\xi_{\rho(s)}}}{\xi_{\rho(s)}}\right] \mathrm{d}s.$$

Proof. Fix $q > 0, t \geq 0$. We first observe that the change of variable $\rho(s) = u$ leads to

$$\int_t^\infty \frac{e^{-\Phi^{-1}(q)\xi_{\rho(s)}}}{\xi_{\rho(s)}}\mathrm{d}s = \int_{\rho(t)}^\infty e^{-\Phi^{-1}(q)\xi_u}\mathrm{d}u.$$

We underline that when $\kappa > 0$ the subordinator ξ reaches ∞ in finite time and that the identity above is valid by using the convention $e^{-\infty} = 0$. Then by Fubini's theorem

$$\int_t^\infty \mathbb{E}\left[\frac{e^{-\Phi^{-1}(q)\xi_{\rho(s)}}}{\xi_{\rho(s)}}\right] \mathrm{d}s = \mathbb{E}\left[\int_{\rho(t)}^\infty e^{-\Phi^{-1}(q)\xi_u}\mathrm{d}u\right].$$

Applying the strong Markov property for Lévy processes to the stopping time $\rho(t)$, we obtain

$$\begin{aligned}\mathbb{E}\left[\int_{\rho(t)}^{\infty} e^{-\Phi^{-1}(q)\xi_u}\mathrm{d}u\right] &= \mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\int_0^{\infty} e^{-\Phi^{-1}(q)\left(\xi_{u+\rho(t)}-\xi_{\rho(t)}\right)}\mathrm{d}u\right]\\ &= \mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right]\int_0^{\infty}\mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_u}\right]\mathrm{d}u.\end{aligned}$$

Observing that $\mathbb{E}[e^{-\Phi^{-1}(q)\xi_t}] = e^{-t\phi(\Phi^{-1}(q))} = e^{-t\Phi'(\Phi^{-1}(q))}$, we conclude that

$$\int_t^{\infty}\mathbb{E}\left[\frac{e^{-\Phi^{-1}(q)\xi_{\rho(s)}}}{\xi_{\rho(s)}}\right]\mathrm{d}s = \frac{1}{\Phi'\circ\Phi^{-1}(q)}\,\mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right] = f(t,q). \qquad \square$$

Proof of Lemma 1.2. Using that for all $a,b>0$, $x\mapsto x^a e^{-bx}$ is bounded on $\mathbb{R}_+$, we obtain easily that the function $q\in(0,\infty)\mapsto f(t,q)$ is infinitely differentiable. Moreover, by Lemma 1.3, for all $t\geq 0, q>0$,

$$\int_t^{\infty}\mathbb{E}\left[\frac{e^{-\Phi^{-1}(q)\xi_{\rho(s)}}}{\xi_{\rho(s)}}\right]\mathrm{d}s = f(t,q) < \infty,$$

therefore $\mathbb{E}[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}/\xi_{\rho(t)}]<\infty$ for almost all $t\geq 0$. Remarking that the function $t\mapsto e^{-\Phi^{-1}(q)\xi_{\rho(t)}}/\xi_{\rho(t)}$ is non-increasing, $t\mapsto\mathbb{E}[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}/\xi_{\rho(t)}]$ is a non-increasing function. As a consequence, $t\in(0,\infty)\mapsto f(t,q)$ is a convex decreasing function and differentiable almost everywhere with

$$\partial_t f(t,q) = -\,\mathbb{E}\left[\frac{e^{-\Phi^{-1}(q)\xi_{\rho(t)}}}{\xi_{\rho(t)}}\right] \quad \text{for almost all } t>0.$$

We then observe that for all $t,q>0$, using Fubini's theorem

$$\begin{aligned}\int_q^{\infty} f(t,r)\mathrm{d}r &= \int_q^{\infty}(\Phi^{-1})'(r)\,\mathbb{E}\left[e^{-\Phi^{-1}(r)\xi_{\rho(t)}}\right]\mathrm{d}r\\ &= \mathbb{E}\left[\int_q^{\infty}(\Phi^{-1})'(r)e^{-\Phi^{-1}(r)\xi_{\rho(t)}}\mathrm{d}r\right] = \mathbb{E}\left[\frac{e^{-\Phi^{-1}(q)\xi_{\rho(t)}}}{\xi_{\rho(t)}}\right].\end{aligned}$$

The function $t\in(0,\infty)\mapsto\mathbb{E}[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}/\xi_{\rho(t)}]$ is therefore continuous, and consequently the function $t\in(0,\infty)\mapsto f(t,q)$ is $\mathcal{C}^1$ for all $q>0$ and

$$\partial_t f(t,q) = -\int_q^{\infty} f(t,r)\mathrm{d}r.$$

Note that this implies that $q\in(0,\infty)\mapsto\partial_t f(t,q)$ is infinitely differentiable. Then, differentiating the above expression with respect to q shows that $\partial_q\partial_t f(t,q) = f(t,q)$.

Similarly, by Fubini's theorem and the change of variables $\rho(s)=u$,

$$\begin{aligned}\int_t^{\infty} f(s,q)\mathrm{d}s &= (\Phi^{-1})'(q)\mathbb{E}\left[\int_t^{\infty} e^{-\Phi^{-1}(q)\xi_{\rho(s)}}\mathrm{d}s\right]\\ &= (\Phi^{-1})'(q)\,\mathbb{E}\left[\int_{\rho(t)}^{\infty}\xi_u e^{-\Phi^{-1}(q)\xi_u}\mathrm{d}u\right]\end{aligned}$$

for $t,q>0$. Moreover, we note that for $u>0$

$$\begin{aligned}&\mathbb{E}\left[\xi_{\rho(t)+u}e^{-\Phi^{-1}(q)\xi_{\rho(t)+u}}\right]\\ &\quad= \mathbb{E}\left[\left(\xi_{\rho(t)}+\left(\xi_{\rho(t)+u}-\xi_{\rho(t)}\right)\right)e^{-\Phi^{-1}(q)\xi_{\rho(t)}}e^{-\Phi^{-1}(q)\left(\xi_{\rho(t)+u}-\xi_{\rho(t)}\right)}\right]\\ &\quad= \mathbb{E}\left[\xi_{\rho(t)}e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right]e^{-u\phi\left(\Phi^{-1}(q)\right)} + \mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right]u\phi'\left(\Phi^{-1}(q)\right)e^{-u\phi(\Phi^{-1}(q))}.\end{aligned}$$

This leads to

$$\begin{aligned}&\int_t^{\infty} f(s,q)\mathrm{d}s\\ &\quad= \frac{\left(\Phi^{-1}\right)'(q)}{\phi\left(\Phi^{-1}(q)\right)}\,\mathbb{E}\left[\xi_{\rho(t)}e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right] + \phi'\left(\Phi^{-1}(q)\right)\frac{\left(\Phi^{-1}\right)'(q)}{\phi\left(\Phi^{-1}(q)\right)^2}\,\mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right]\\ &\quad= -\partial_q f(t,q),\end{aligned}$$

since by definition of f

$$\partial_q f(t,q) = (\Phi^{-1})''(q)\,\mathbb{E}\left[e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right] - (\Phi^{-1})'(q)^2\,\mathbb{E}\left[\xi_{\rho(t)}e^{-\Phi^{-1}(q)\xi_{\rho(t)}}\right]$$

and $(\Phi^{-1})' = 1/\phi(\Phi^{-1})$. Differentiating with respect to t shows that we also have $\partial_t\partial_q f(t,q) = f(t,q)$.

The continuity at 0 of $t \mapsto f(t,q)$ is obvious, for all $q > 0$, by using the expression of f as an integral as noticed in Lemma 1.3. It remains to identify the initial expression for $f(0,q)$ in terms of the measure L. Since $\xi_{\rho(0)} = 0$ when $c > 0$ or π is infinite, and $\xi_{\rho(0)}$ is the first jump time of ξ otherwise, one has

$$\mathbb{E}\left[e^{-q\xi_{\rho(0)}}\right] = \begin{cases} 1 & \text{when } c > 0 \text{ or } \pi \text{ is infinite} \\ \frac{1}{\kappa+\pi(0,\infty)}\int_0^\infty e^{-qx}\pi(\mathrm{d}x) & \text{when } c = 0 \text{ and } \pi \text{ is finite.} \end{cases}$$

When $c > 0$ or π is infinite, recalling (1.3), this indeed yields

$$f(0,q) = (\Phi^{-1})'(q) = \int_0^\infty xe^{-qx}L(\mathrm{d}x).$$

When $c = 0$ and π is finite, we have

$$f(0,q) = (\Phi^{-1})'(q)\frac{1}{\kappa+\pi(0,\infty)}\int_0^\infty e^{-\Phi^{-1}(q)x}\pi(\mathrm{d}x).$$

Using that in this case $\Phi(q) = (\kappa+\pi(0,\infty))q - \int_0^\infty(1-e^{-qx})\pi(\mathrm{d}x)/x$, we see that

$$q = (\kappa+\pi(0,\infty))\Phi^{-1}(q) - \int_0^\infty\left(1-e^{-\Phi^{-1}(q)x}\right)\pi(\mathrm{d}x)/x,$$

which, by differentiating, leads us to

$$f(0,q) = (\Phi^{-1})'(q)\frac{1}{\kappa+\pi(0,\infty)}\int_0^\infty e^{-\Phi^{-1}(q)x}\pi(\mathrm{d}x) = (\Phi^{-1})'(q) - \frac{1}{\kappa+\pi(0,\infty)}.$$

By (1.3) and since $c = 0$, this last expression is equal to $\int_0^\infty xe^{-qx}L(\mathrm{d}x)$ as expected. □

Solving the partial differential equation of Lemma 1.2, we obtain an expression for the Laplace transform of the function $t \mapsto f(t,q)$ for each $q > 0$. This leads to an explicit expression of $f(t,q)$ in terms of the measure L, which proves Theorem 0.1.

Lemma 1.4. *For $\lambda > 0$, we write $F_\lambda : q \in (0,\infty) \mapsto \int_0^\infty e^{-\lambda t}f(t,q)\mathrm{d}t$. Then for all fixed $q > 0$, we have*

$$F_\lambda(q) = \int_0^\infty\left(\int_0^\infty e^{-qx-\frac{t}{x}}xL(\mathrm{d}x)\right)e^{-\lambda t}\mathrm{d}t, \quad \forall\,\lambda > 0 \tag{1.6}$$

which implies that $f(t,q) = \int_0^\infty e^{-qx-\frac{t}{x}}xL(\mathrm{d}x)$ for all $t \geq 0, q > 0$. In particular, f is infinitely differentiable on $(0,\infty)^2$.

Proof. Fix $\lambda > 0$ and note that $F_\lambda(q)$ is finite for all $q > 0$, since the function $t \in (0,\infty) \mapsto f(t,q)$ is bounded. An integration by part gives

$$\begin{aligned} F_\lambda(q) =& \left[-\frac{e^{-\lambda t}}{\lambda}f(t,q)\right]_{t=0}^{t=\infty} + \frac{1}{\lambda}\int_0^\infty e^{-\lambda t}\partial_t f(t,q)\mathrm{d}t \\ =& \frac{1}{\lambda}\int_0^\infty xe^{-qx}L(\mathrm{d}x) + \frac{1}{\lambda}\int_0^\infty e^{-\lambda t}\partial_t f(t,q)\mathrm{d}t, \end{aligned}$$

using that $e^{-\lambda t}f(t,q) \to 0$ as $t \to \infty$, and that $f(t,q) \to \int_0^\infty xe^{-qx}L(\mathrm{d}x)$ as $t \to 0$ by Lemma 1.2. We then have, for all $q > 0$,

$$\begin{aligned} \partial_q F_\lambda(q) &= \frac{-1}{\lambda}\int_0^\infty x^2e^{-qx}L(\mathrm{d}x) + \frac{1}{\lambda}\int_0^\infty e^{-\lambda t}\partial_q\partial_t f(t,q)\mathrm{d}t \\ &= \frac{-1}{\lambda}\int_0^\infty x^2e^{-qx}L(\mathrm{d}x) + \frac{1}{\lambda}F_\lambda(q), \end{aligned}$$

using again Lemma 1.2 (we can apply Lebesgue's dominated convergence theorem since, for all $q_0 > 0$ and $t > 0$,

$$\sup_{q\geq q_0}\left|e^{-\lambda t}\partial_q\partial_t f(t,q)\right| = \sup_{q\geq q_0}\left|e^{-\lambda t}f(t,q)\right| = e^{-\lambda t}f(t,q_0),$$

since $(\Phi^{-1})'$ is decreasing and Φ^{-1} is increasing).

The function $q \in (0,\infty) \mapsto F_\lambda(q)$ is therefore a solution to a linear differential equation of the first order. By standard arguments, it follows that

$$F_\lambda(q) = e^{\frac{q}{\lambda}}\int_q^\infty e^{-\frac{u}{\lambda}}\left(\frac{1}{\lambda}\int_0^\infty x^2e^{-ux}L(\mathrm{d}x)\right)\mathrm{d}u.$$

Using Fubini's theorem and the change of variable $t = (u-q)x/\lambda$, we then get the expression (1.6) for all $q, \lambda > 0$. The final equation follows from Laplace inversion theorem and the fact that the functions $t \mapsto f(t,q)$ and $t \mapsto \int_0^\infty e^{-qx-\frac{t}{x}} xL(\mathrm{d}x)$ are continuous on $(0,\infty)$. □

Proof of Theorem 0.1. Observe that the definition (1.4) of f implies that

$$f(\Phi(q),t) = (\Phi^{-1})'(\Phi(q))\,\mathbb{E}\left[e^{-q\xi_{\rho(t)}}\right] = \frac{1}{\phi(q)}\,\mathbb{E}\left[e^{-q\xi_{\rho(t)}}\right].$$

As a result, Lemma 1.4 yields

$$\mathbb{E}\left[e^{-q\xi_{\rho(t)}}\right] = \phi(q) f\left(t,\Phi(q)\right) = \phi(q)\int_0^\infty e^{-\Phi(q)x-\frac{t}{x}}\; xL(\mathrm{d}x), \quad \forall\, t \geq 0, q > 0,$$

which completes the proof. □

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